Question cum Answer booklet

PH 102, Physics-II
Supplementary Examination
Department of Physics, IIT Guwahati
Total marks: 40; Duration: 3 Hours

Please read the instructions carefully

1. Please write your name and roll number on each and every page.
2. All questions are compulsory. The details of the calculation need to be shown by providing the intermediate steps.
3. Please write the answer to each question exactly in the space provided against that question. Do not exceed the space limit. Answers written elsewhere will not be considered for evaluation.
4. Use the additional answer booklet provided for doing the rough work.
5. At the end of the exam, please return this Question cum Answer booklet along with the other booklet used for your rough work.

Name of the student :

Roll number of the student :

Signature of the student

Signature of the Invigilator

Signature of the Examiner

 Name: **Roll No:**

Useful formulae:

Cartesian. $d\mathbf{l} = dx\,\hat{\mathbf{x}} + dy\,\hat{\mathbf{y}} + dz\,\hat{\mathbf{z}}$; $\quad d\tau = dx\,dy\,dz$

Gradient: $$\nabla t = \frac{\partial t}{\partial x}\hat{\mathbf{x}} + \frac{\partial t}{\partial y}\hat{\mathbf{y}} + \frac{\partial t}{\partial z}\hat{\mathbf{z}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Curl: $$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\mathbf{z}}$$

Laplacian: $$\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$$

Spherical. $d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$; $\quad d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi$

Gradient: $$\nabla t = \frac{\partial t}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial t}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}$$

Curl: $$\nabla \times \mathbf{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{\mathbf{r}}$$
$$+ \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\boldsymbol{\phi}}$$

Laplacian: $$\nabla^2 t = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial t}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial \phi^2}$$

Cylindrical. $d\mathbf{l} = ds\,\hat{\mathbf{s}} + s\,d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\mathbf{z}}$; $\quad d\tau = s\,ds\,d\phi\,dz$

Gradient: $$\nabla t = \frac{\partial t}{\partial s}\hat{\mathbf{s}} + \frac{1}{s}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial t}{\partial z}\hat{\mathbf{z}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

Curl: $$\nabla \times \mathbf{v} = \left[\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right]\hat{\mathbf{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{\boldsymbol{\phi}} + \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi}\right]\hat{\mathbf{z}}$$

Laplacian: $$\nabla^2 t = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial t}{\partial s}\right) + \frac{1}{s^2}\frac{\partial^2 t}{\partial \phi^2} + \frac{\partial^2 t}{\partial z^2}$$

Space for providing the solution for Question-1 only.

1. Consider the upper half surface of the sphere $x^2 + y^2 + z^2 = 4$. For the given vector field $\vec{A} = (2x - y)\hat{i} - 2yz^2\hat{j} - 2zy^2\hat{k}$
   (a) Calculate the surface integral of the curl of the vector field over this surface [2 Marks]
   (b) Calculate the closed line integral and verify Stoke's theorem. [3 Marks]

(a) Surface integral : $\int_S (\vec{\nabla} \times \vec{A}) \cdot \vec{ds}$

$$\nabla \times \vec{A} = \hat{x}\left[\frac{\partial(-2y^2z)}{\partial y} - \frac{\partial}{\partial z}(-2yz^2)\right] + \hat{y}\left[\frac{\partial(2x-y)}{\partial z} - \frac{\partial}{\partial x}(-2zy^2)\right]$$
$$+ \hat{z}\left[\frac{\partial(-2zy^2)}{\partial x} - \frac{\partial}{\partial y}(2x-y)\right]$$
$$= \hat{x}(-4yz + 4yz) + \hat{y}(0-0) + \hat{z}(0+1) = \hat{z}.$$ [1]

Surface integral: $\vec{ds} = \hat{r}\, r^2 \sin\theta\, d\theta\, d\phi$.

$$\int_S \vec{\nabla}\times\vec{A} \cdot ds = \int \hat{z}\cdot\hat{r}\, r^2 \sin\theta\, d\theta\, d\phi$$
$$= r^2 \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi/2} \cos\theta \sin\theta\, d\theta\, d\phi$$
$$= 2\pi (2)^2 \int_0^{\pi/2} \cos\theta \sin\theta\, d\theta$$
$$= 8\pi \frac{1}{2}\int_0^{\pi/2} \sin 2\theta\, d\theta$$
$$= 4\pi\left[-\frac{\cos 2\theta}{2}\right]_0^{\pi/2} = 4\pi$$ [1]

(b) Line integral: $\vec{dl} = \hat{x}\, dx + \hat{y}\, dy$

using $z=0$ on C (xy plane)

$$\oint_C \vec{A}\cdot\vec{dl} = \oint_C \hat{x}\,(2x-y)\cdot[\hat{x}\,dx + \hat{y}\,dy]$$

Thus; $\oint_C \vec{A}\cdot\vec{dl} = \oint_C (2x-y)\,dx$ [1]

Using cylindrical coordinates:

$$x = r\cos\phi = 2\cos\phi \ , \ y = 2\sin\phi \, .$$ [1]

$$\therefore \quad \frac{dx}{d\phi} = -r\sin\phi \Rightarrow dx = -2\sin\phi\, d\phi$$

$$= \int_0^{2\pi} [2\,(2\cos\phi) - 2\sin\phi](-2\sin\phi)\, d\phi$$

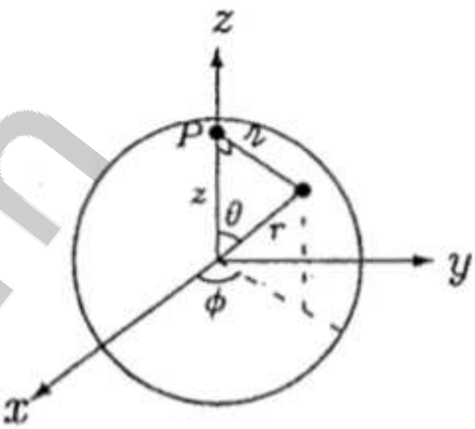
$$= -8\int_0^{2\pi} \cos\phi\sin\phi\, d\phi + 4\int_0^{2\pi} \sin^2\phi\, d\phi$$

$$= 0 + 4\pi = 4\pi$$ [1]

Space for providing the solution for Question-2 only.

2. Calculate the electrostatic potential inside a uniformly charged solid sphere of radius R and total charge q. [5 Marks]

Rotate the $\vec{OP}$ such that the point P lies on the z-axis.



$$V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho}{\mathscr{r}}\, d\tau$$

where ρ is constant

$$d\tau = r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$\mathscr{r} = \sqrt{z^2 + r^2 - 2rz\cos\theta}$$

$$\therefore\quad V = \frac{\rho}{4\pi\epsilon_0}\int \frac{r^2 \sin\theta\, dr\, d\theta\, d\phi}{\sqrt{z^2 + r^2 - 2rz\cos\theta}}$$ [2]

Now: $$\int_0^{2\pi} d\phi = 2\pi$$

$$\int_0^{\pi} \frac{\sin\theta}{\sqrt{z^2 + r^2 - 2rz\cos\theta}}\, d\theta = \frac{1}{rz}\left(\sqrt{r^2 + z^2 - 2rz\cos\theta}\right)\Big|_0^{\pi}$$

$$= \frac{1}{rz}\left(\sqrt{r^2 + z^2 + 2rz} - \sqrt{r^2 + z^2 - 2rz}\right)$$

$$= \frac{1}{rz}\left(r + z - |r - z|\right)$$

$$= \begin{cases} 2/z & \text{if } r < z \\ 2/r & \text{if } r > z \end{cases}$$ [1]

$$\therefore \quad V = \frac{\rho}{4\pi\epsilon_0} \cdot 2\pi \cdot 2 \left\{ \int_0^z \frac{1}{z} r^2 dr + \int_z^R \frac{1}{r} r^2 dr \right\}$$

$$= \frac{\rho}{\epsilon_0} \left\{ \frac{1}{z} \frac{z^3}{3} + \frac{R^2 - z^2}{2} \right\} = \frac{\rho}{2\epsilon_0} \left(R^2 - \frac{z^2}{3} \right)$$

[1]

But $\rho = \dfrac{q}{\frac{4}{3}\pi R^3}$,

so $$V(z) = \frac{1}{2\epsilon_0} \frac{3q}{4\pi R^3} \left(R^2 - \frac{z^2}{3} \right)$$

$$= \frac{q}{8\pi\epsilon_0 R} \left(3 - \frac{z^2}{R^2} \right)$$

[1]

$$\therefore \quad V(r) = \frac{q}{8\pi\epsilon_0 R} \left(3 - \frac{r^2}{R^2} \right)$$

Space for providing the solution for Question-3 only.

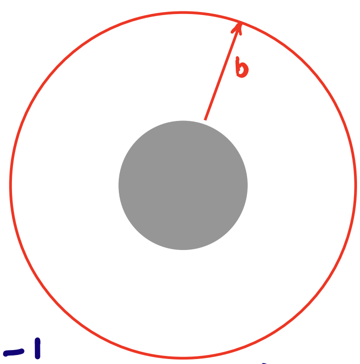
3. Calculate the capacitance of Earth assuming it to be a conducting sphere of radius $R = 6400\text{km}$. [5 Marks]

Solution:

A typical spherical capacitor is shown in the figure below. It consists of a solid inner conducting sphere of radius a, surrounded by outer conducting shell of radius b. The inner sphere carries +q and the outer carries −q. So, the electric field is non-zero only between the spheres and is

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}$$ [1]

The potential difference between the spheres is



$$\Delta V = -\int_b^a \vec{E}\cdot\vec{dl} = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

[1]

$$\therefore \quad C = \frac{Q}{\Delta V} = 4\pi\epsilon_0\left(\frac{1}{a}-\frac{1}{b}\right)^{-1} \quad \text{— } (1)$$ [1]

We can think of earth as a spherical capacitor with inner sphere of radius $a = R_0$ and outer conducting sphere can be assumed to be at infinity.

we assume that a charge is uniformly distributed over the surface.

From eq^n. (1)

$$C = \frac{Q}{\Delta V} = 4\pi\epsilon_0 \left(\frac{1}{a} - \frac{1}{b}\right)^{-1}$$

$a = R_0$
$b \to \infty$

$$= 4\pi\epsilon_0 \left(\frac{1}{R_0} - \frac{1}{\infty}\right)^{-1}$$

[1]

$$= 4\pi\epsilon_0 R_0$$

Given $R_0 = 6400$ km

$$C = 7.12 \times 10^{-4}\ F.$$

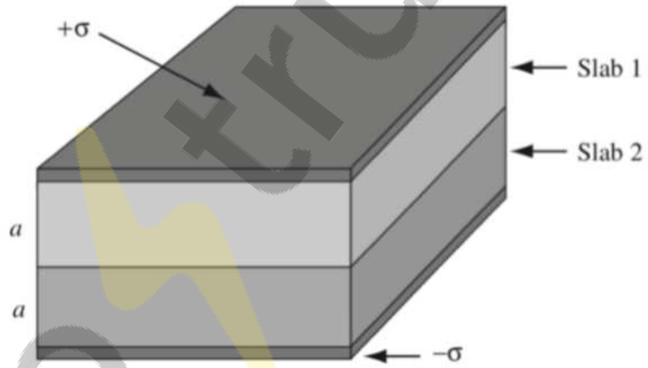
$$= 712\ \mu F$$

[1]

 Name: **Roll No:**

Space for providing the solution for Question-4 only.

4. The space between the plates of a parallel-plate capacitor (as shown in the figure) is filled with two slabs of different dielectric materials. Each slab has thickness a. The dielectric constant of slab-1 is $k_1 = 2$ and that of the slab-2 is $k_2 = 1.5$. The free surface charge density on the top place is σ while that on the bottom plate is $-\sigma$.
   (a) Find the electric displacement in each slab. [1 Mark]
   (b) Find the electric field in each slab. [1 Mark]
   (c) Find the polarization in each slab. [2 Marks]
   (d) Find the potential difference between two plates. [1 Mark]



Solution:

(a) Apply $\oint \vec{D}\cdot d\vec{a} = Q_{f\,enc.}$ to the Gaussian surface shown

$DA = \sigma A$

$\Rightarrow \quad D = \sigma$

(Note: $D=0$ inside the metal plate)

This is true in both slabs [1]
$\vec{D}$ points down in both the slabs.

(b) $\vec{D} = \epsilon \vec{E} \Rightarrow E = \sigma / \epsilon_1$ in slab 1

$E = \sigma / \epsilon_2$ in slab 2

But $\epsilon = \epsilon_0 \epsilon_r$, so $\epsilon_1 = 2\epsilon_0$; $\epsilon_2 = \frac{3}{2}\epsilon_0$

$E_1 = \sigma / 2\epsilon_0$; $\vec{E_2} = \dfrac{2\sigma}{3\epsilon_0}$ [1]

(c) $\vec{P} = \epsilon_0 \chi_e \vec{E}$

so $P = \epsilon_0 \chi_e \sigma / (\epsilon_0 \epsilon_r)$

$= (\chi_e / \epsilon_r)\, \sigma$;

$\chi_e = \epsilon_r - 1$

$\Rightarrow P = (1 - \epsilon_r^{-1})\, \sigma$

$\therefore P_1 = \sigma/2$ and $P_2 = \sigma/3$ [2]

(d) $V = E_1 a + E_2 a = (\sigma a / 6\epsilon_0)(3+4)$

$= \dfrac{7\sigma a}{6\epsilon_0}$ [1]

Q.5 A current distribution produces the vector potential

$$\vec{A}(r,\theta,\phi)=\hat{\phi}\frac{\mu_0}{4\pi}\frac{A_0\sin\theta}{r}e^{-\lambda r}$$

What is the magnetic dipole moment associated with this current distribution?

[4]

Solution: To find magnetic dipole moment, let us calculate $\vec{B}$ and $\vec{J}$ first:

$$\vec{B}=\vec{\nabla}\times\vec{A}=\frac{\mu_0 A_0}{4\pi}\left[\hat{r}\frac{2\cos\theta}{r^2}+\hat{\theta}\frac{\lambda\sin\theta}{r}\right]e^{-\lambda r} \qquad [+1]$$

Since $\vec{\nabla}\times\vec{B}=\mu_0\vec{J}$, therefore,

$$\vec{J}=\frac{1}{\mu_0}(\vec{\nabla}\times\vec{B})=\hat{\phi}\frac{A_0}{4\pi}\sin\theta\left\{\frac{2}{r^3}-\frac{\lambda^2}{r}\right\}e^{-\lambda r} \qquad [+1]$$

Therefore, magnetic dipole moment is,

$$\vec{m}=\frac{1}{2}\int\left(\vec{r}\times\vec{J}\right)d^3r$$

$$=-\frac{A_0}{8\pi}\int d^3r\hat{\theta}r\sin\theta\left\{\frac{2}{r^3}-\frac{\lambda^2}{r}\right\}e^{-\lambda r} \qquad [+1]$$

Now since, $\hat{\theta}=\hat{i}\cos\theta\cos\phi+\hat{j}\cos\theta\sin\phi-\hat{k}\sin\theta$, we get,

$$\vec{m}=+\frac{A_0}{8\pi}\hat{k}\int r^3\sin^3\theta\left(\frac{2}{r^3}-\frac{\lambda^2}{r}\right)e^{-\lambda r}drd\theta$$

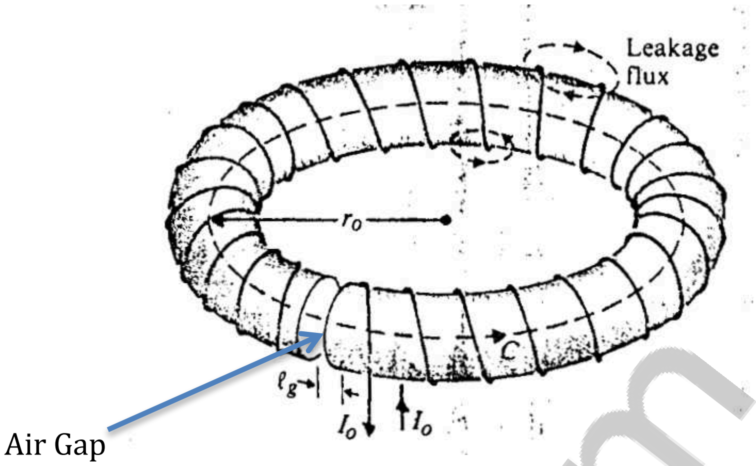
$$=\frac{A_0}{8\pi}\hat{k}\frac{4}{3}\left(\frac{2}{\lambda}-\frac{2}{\lambda}\right)=0 \qquad [+1]$$

Q.6 Assume that N turns of wire are wound around a toroidal core of a ferromagnetic material with permeability μ. The core has a mean radius r_0, a circular cross section of radius $a\left(a<<r_0\right)$, and a narrow air gap of length l_g as shown in the figure. A steady current I_0 flows in the wire. Ignore flux leakage through the toroidal surface and the fringing effect of the flux in the air gap. Determine:

(a) the magnetic flux density $\vec{B}_f$, in the ferromagnetic core; [2]

(b) the magnetic field intensity, $\vec{H}_f$, in the core; and [2]

(c) the magnetic field intensity, $\vec{H}_g$, in the air gap. [2]



Solution:

a) Applying Ampere's circuital law, around the circular contour C, which has a mean radius r_0, we have

$$\oint_C \vec{H}.dl = NI_0 \qquad \text{-------(1)}$$

If the flux leakage is neglected, the same total flux will flow in both the ferromagnetic core and the air gap. If the fringing effect of the flux in the air gap is also neglected, the magnetic flux density $\vec{B}$ in both the core and the air gap will also be the same. However, because of the different permeability, the magnetic field intensities in both parts will be different. We have,

$$\vec{B}_f = \vec{B}_g = \hat{\phi} B_f \qquad \text{--------(2)} \quad [+1]$$

In the ferromagnetic core,

$$\vec{H}_f = \hat{\phi}\frac{B_f}{\mu} \qquad \text{--------(3)}$$

and in the air gap,

$$\vec{H}_g = \hat{\phi}\frac{B_f}{\mu_0} \qquad \text{--------(4)}$$

Substituing Eqs. (2), (3), and (4) in Eq. (1), we have,

$$\frac{B_f}{\mu}\left(2\pi r_0 - l_g\right) + \frac{B_f}{\mu_0} l_g = NI_0$$

and therefore,

$$\vec{B}_f = \hat{\phi}\frac{\mu_0 \mu N I_0}{\mu_0\left(2\pi r_0 - l_g\right) + \mu l_g} \qquad \text{--------(5)} \qquad [+1]$$

b) From Eqs. (3) and (5), we get,

$$\vec{H}_f = \hat{\phi}\frac{\mu_0 N I_0}{\mu_0\left(2\pi r_0 - l_g\right) + \mu l_g} \qquad \text{---------(6)} \qquad [+2]$$

c) Similarly, from Eqs. (4) and (5), we get,

$$\vec{H}_g = \hat{\phi}\frac{\mu N I_0}{\mu_0\left(2\pi r_0 - l_g\right) + \mu l_g} \qquad \text{---------(7)} \qquad [+2]$$

Since $H_g / H_f = \mu / \mu_0$, the magnetic field intensity in the air gap is much stronger that that in the ferromagnetic core.

Q.7. Consider a solid cylindrical wire of radius R1 surrounded by a thin long cylindrical coaxial shell of radius R2. In the inner cylindrical solid wire, current I is distributed uniformly. In the outer cylindrical shell the same current flows, but in the opposite direction. Find the

(a) Magnetic energy stored in the cable per unit length of the cable. [3]

(b) Self inductance per unit length of the cable. [1]

Solution:

(a) To find the magnetic energy stored, one has to find the magnetic field in all the regions, using Ampere's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

In the inner solid cylinder of radius R_1, the current I is distributed uniformly and hence for a loop of radius $s < R_1$, the enclosed current is $I_{\text{enc}} = Is^2/R_1^2$. The enclosed current in the intermediate region between two wires $R_1 < s < R_2$ is same as I. The enclosed current for $s > R_2$ on the other hand, is zero as equal and opposite currents flow in the two wires. Using these, we can find the magnetic field as

$$\vec{B} = \frac{\mu_0 I s}{2\pi R_1^2}\hat{\phi}, \; s < R_1$$

[1]

$$\vec{B} = \frac{\mu_0 I}{2\pi s}\hat{\phi}, \; R_1 < s < R_2$$

$$\vec{B} = 0, \; s > R_2$$

Thus the stored magnetic energy is

$$E_{\text{mag}} = \frac{1}{2\mu_0}\int B^2 d\tau = \frac{1}{2\mu_0}\left[\int_0^{R_1}\left(\frac{\mu_0 I s}{2\pi R_1^2}\right)^2 s ds d\phi dz + \int_{R_1}^{R_2}\left(\frac{\mu_0 I}{2\pi s}\right)^2 s ds d\phi dz\right]$$

[1]

After integrating the angular coordinate between $0 - 2\pi$, we can find the magnetic energy per unit length l as

$$\frac{E_{\text{mag}}}{l} = \frac{\mu_0 I^2}{4\pi}\left(\frac{1}{4} + \ln\frac{R_2}{R_1}\right)$$

[1]

(b) Self inductance L is related to the stored energy as $E_{\text{mag}} = LI^2/2$. Therefore, self inductance per unit length is

$$L/l = \frac{2\mu_0}{4\pi}\left(\frac{1}{4} + \ln\frac{R_2}{R_1}\right)$$

[1]

Q.8. A plane EM wave of frequency ω, travelling in air along the z-direction and polarized along the x-direction in incident normally on a flat surface (a medium of refractive index n). Take the incident, reflected and transmitted wave amplitudes as E_{0I}, E_{0R}, and E_{0T}, respectively.

(a) Using EM boundary conditions at the interface, find the relations between the incident, reflected and transmitted wave amplitudes. Take $\mu_1 = \mu_2 = 1$. [2]

(b) Using the above relations, find the reflection and transmission coefficients of the wave in terms of n. [3]

(c) Find the corresponding reflection coefficient of diamond (take n=2.4). [1]

SOLUTION: 8(a):

$$\tilde{E}_{0_I} + \tilde{E}_{0_R} = \tilde{E}_{0_T}, \qquad (9.78)$$ [1]

$$\frac{1}{\mu_1}\left(\frac{1}{v_1}\tilde{E}_{0_I} - \frac{1}{v_1}\tilde{E}_{0_R}\right) = \frac{1}{\mu_2}\left(\frac{1}{v_2}\tilde{E}_{0_T}\right), \qquad (9.79)$$ [1]

$$\tilde{E}_{0_I} - \tilde{E}_{0_R} = \beta\tilde{E}_{0_T}, \qquad (9.80)$$

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}. \qquad (9.81)$$

(b)

$$\tilde{E}_{0_R} = \left(\frac{1-\beta}{1+\beta}\right)\tilde{E}_{0_I}, \quad \tilde{E}_{0_T} = \left(\frac{2}{1+\beta}\right)\tilde{E}_{0_I}. \qquad (9.82)$$ [1]

These results are strikingly similar to the ones for waves on a string. Indeed, if the rmittivities μ are close to their values in vacuum (as, remember, they *are* for most media), en $\beta = v_1/v_2$, and we have

$$\tilde{E}_{0_R} = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)\tilde{E}_{0_I}, \quad \tilde{E}_{0_T} = \left(\frac{2v_2}{v_2 + v_1}\right)\tilde{E}_{0_I}, \qquad (9.83)$$

hich are *identical* to Eqs. 9.30. In that case, as before, the reflected wave is *in phase* (right le up) if $v_2 > v_1$ and *out of phase* (upside down) if $v_2 < v_1$; the real amplitudes are lated by

$$E_{0_R} = \left|\frac{v_2 - v_1}{v_2 + v_1}\right| E_{0_I}, \quad E_{0_T} = \left(\frac{2v_2}{v_2 + v_1}\right) E_{0_I}, \qquad (9.84)$$

, in terms of the indices of refraction,

$$E_{0_R} = \left|\frac{n_1 - n_2}{n_1 + n_2}\right| E_{0_I}, \quad E_{0_T} = \left(\frac{2n_1}{n_1 + n_2}\right) E_{0_I}. \qquad (9.85)$$

$$R \equiv \frac{I_R}{I_I} = \left(\frac{E_{0_R}}{E_{0_I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \boxed{= \left(\frac{\boldsymbol{n-1}}{\boldsymbol{n+1}}\right)^2}$$ [1]

the transmitted intensity to the incident intensity is

$$T \equiv \frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{E_{0_T}}{E_{0_I}}\right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2}. \qquad \boxed{\cancel{= \left(\frac{\boldsymbol{4n}}{\boldsymbol{n+1}}\right)^2}}$$ [1]

(c) R = [(2.4-1)/(2.4+1)]² = 0.17 [1]

(4n)/(n+1)^2